

Al-Aqsa University
Faculty of Engineering
Intelligence Systems &
Computer Engineering Dept.



جامعة الأقصى
كلية الهندسة
قسم هندسة النظم الذكية و الحاسوب

Smartphones Programming (ENGS4301)

Assignment 2: Dart Project

Presented to: Eng. Ibrahim Shehada

Student name: Samar M. Balousha

Student number: 2320222601

[Link to Github](#)

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Dart Programming — OOP (Classes, Inheritance)

Code:

```
// Shopping Cart and Checkout System in Dart

// item class to represent a product in the shopping cart
class Item {
  int id;
  String name;
  double price ;
  int qty ;

  Item(this.id, this.name, this.price, this.qty);

  void printItem(){
    print('ID: $id || Name: $name || Price: \${price.toStringAsFixed(2)} || Quan-
tity: $qty');
  }
}

// cart class to manage the shopping cart

class Cart {
  List<Item> items =[];
  void addItem(Item item){
    for (var i in items){
      if (i.id == item.id){
        i.qty += item.qty;
        return;
      }
    }
    items.add(item);
    print('Item added to cart: ${item.name}');
  }

  // remove an item method : remove one item , it quantity = 0 remove the item
  void removeItem(int id ){
    for (int i = 0; i< items.length ; i++){
      if(items[i].id == id ){

        if (items[i].qty > 0){
          items[i].qty -= 1;
        }
        if (items[i].qty == 0){
          items.removeAt(i);
        }
      }
    }
  }
}
```

```

break;
    }

}
}

// subtotal method : sum of (price * quantity) for all items

double subtotal(){
    double total = 0.0;
    for (var item in items){
        total += item.price * item.qty;
    }
    return total;
}

// print all items and the subtotal
void printCart(){
    print('Shopping Cart');
    for (var item in items){
        item.printItem();
    }
    print('Subtotal: \${subtotal().toStringAsFixed(2)}');
}

// clear cart method : remove all items from the cart
void clearCart(){
    items.clear();
    print('Cart cleared');
}

}

// checkout Class
class Checkout extends Cart {

    double deliveryFee = 10.0; // fixed delivery fee

    // tax = 10% og subtotal
    double get tax => subtotal() * 0.10;

    // total = subtotal + tax + delivery fee
    double get total => subtotal() + tax + deliveryFee;

    // print full invoice method : items , subtotal, tax, delivey fees, total

```

```
void printInvoice(){
    print('--- Invoice ---');
    for (var item in items){
        item.printItem();
    }
    print('Subtotal: \${subtotal().toFixed(2)}');
    print('Tax (10%): \${tax.toFixed(2)}');
    print('Delivery Fee: \${deliveryFee.toFixed(2)}');
    print('Total: \${total.toFixed(2)}');
}
```

```
}
```

```
// Main Method :
```

```
void main(){
```

```
    Item Shirt = Item(1, 'T-Shirt', 50.0, 1);
```

```
    Item Jeans = Item(2, 'Jeans', 80, 2);
```

```
    Item Scarf = Item(3, 'Scarf', 30, 1);
```

```
    Checkout checkout = Checkout();
```

```
    checkout.addItem(Shirt);
```

```
    checkout.addItem(Jeans);
```

```
    checkout.addItem(Scarf);
```

```
        checkout.addItem(Shirt); // increase quantity of shirt to 2
```

```
    checkout.printCart();
```

```
    checkout.removeItem(3);
```

```
    print(" after remove ");
```

```
    checkout.printInvoice();
```

```
}
```

Result:

Item added to cart: T-Shirt

Item added to cart: Jeans

Item added to cart: Scarf

Shopping Cart

ID: 1 || Name: T-Shirt || Price: \$50.00 || Quantity: 2

ID: 2 || Name: Jeans || Price: \$80.00 || Quantity: 2

ID: 3 || Name: Scarf || Price: \$30.00 || Quantity: 1

Subtotal: \$290.00

after remove

--- Invoice ---

ID: 1 || Name: T-Shirt || Price: \$50.00 || Quantity: 2

ID: 2 || Name: Jeans || Price: \$80.00 || Quantity: 2

Subtotal: \$260.00

Tax (10%): \$26.00

Delivery Fee: \$10.00

Total: \$296.00

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